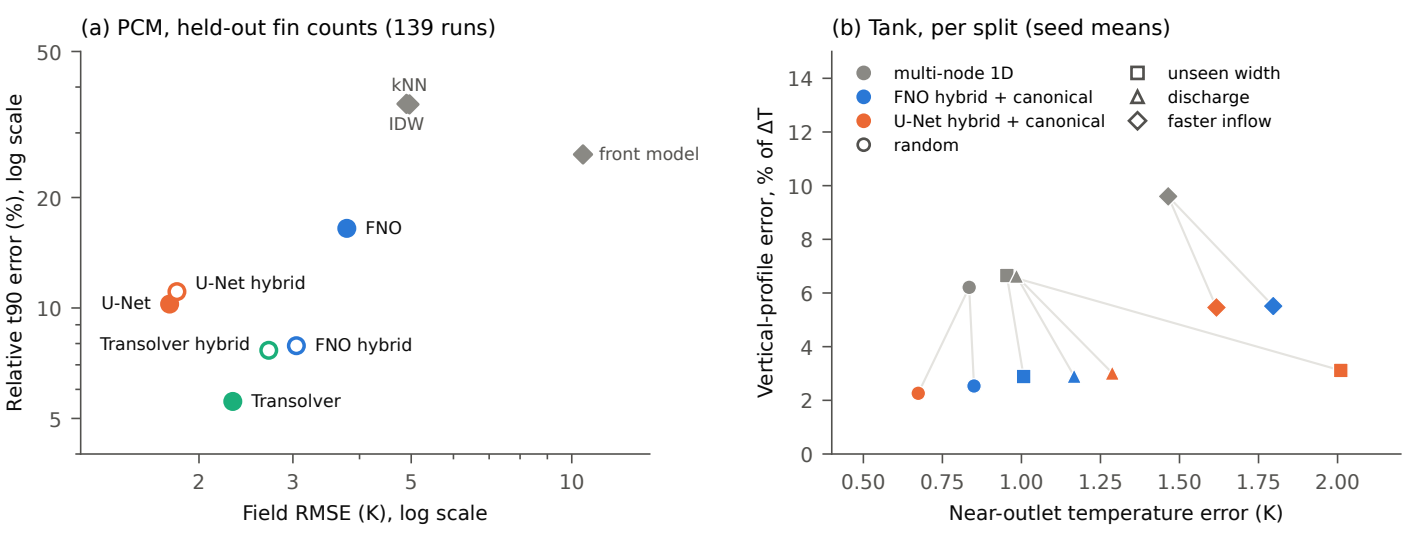
(a) PCM, held-out fin counts (139 runs)
Relative t90 error (%), log scale
Field RMSE (K), log scale
50
20
10
5
2
3
5
10
kNN
IDW
front model
FNO
U-Net hybrid
U-Net
Transolver hybrid
FNO hybrid
Transolver
(b) Tank, per split (seed means)
Vertical-profile error, % of ΔT
Near-outlet temperature error (K)
multi-node 1D
FNO hybrid + canonical
U-Net hybrid + canonical
random
unseen width
discharge
faster inflow
0
2
4
6
8
10
12
14
0.50
0.75
1.00
1.25
1.50
1.75
2.00